ECI v7.4 amendment (v2): τ as a CM-anchored attractor, the Damerell–ladder Cardy/CKM bridge H_7', and a CSD$(1+\sqrt{6})$ Littlest Modular Seesaw at $K = \mathbb{Q}(i)$

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Abstract

The v7.3 axiomatic system of the Extended Cosmological Index (ECI) research programme was archived in version 6.0.53 of the project repository (Zenodo DOI `10.5281/zenodo.20030684`) on 2026-05-05. An internal Lakatos-style audit (G1.15) identified one refuted axiom: H_7 asserted a Damerell–Coates–Sinnott bridge for the LMFDB CM newform $f_{4.5.\mathrm{b.a}}$ (CM by $\mathbb{Q}(i)$, weight $k = 5$) at the central half-integer $s = k/2 = 5/2$, which lies *outside* the Damerell/Shimura/Deligne algebraicity domain for odd k.

This v2 note records the v7.3 $\to$ v7.4 amendment, expanded with six wave 4-5 follow-ups (A14, A16, A17, A22, A24, A26). Three independent sub-agents (H7-A, H7-B, H7-C) explored rescue paths and converged on keeping $f_{4.5.\mathrm{b.a}}$ as the v7.4 CM anchor. We propose two replacements. First, H_5' relaxes "τ is the strict S-fixed point i" to "τ is a CM-anchored *attractor*, $|\tau - i| \lesssim 0.2$", supported empirically by a 30×30 χ^2 scan over seven fermion observables (W1, PC compute, 2026-05-05) which locates a basin at $\tau^* = -0.1897 + 1.0034\,i$ with $\chi^2/\mathrm{dof} = 1.05$ and $|\tau^* - i| = 0.19$ – a factor ≈ 3 closer to i than the LYD20 published best fit [**?**]. Second, H_7' replaces the half-integer Damerell–CS bridge by an INTEGER-point Damerell ladder

$$\boxed{L(f_{4.5.\mathrm{b.a}}, m)\,\pi^{4-m}/\Omega_K^4 \;\in\; \{1/10,\, 1/12,\, 1/24,\, 1/60\} \;\text{ for }\; m \in \{1,2,3,4\}}$$

which is the standard Eisenstein–Kronecker ladder for a weight-5 CM-by-$\mathbb{Q}(i)$ newform. The same ladder rationals admit two *numerological* CKM alignments (A17): $|V_{us}| = (9/4)\,\alpha_1 = 9/40 = 0.22500$ matches PDG $0.22501(68)$ to 0.015σ, and $|V_{cb}|^2 = \alpha_1\,\alpha_4 = 1/600 = 1.667 \times 10^{-3}$ matches HFLAV $(1.668 \pm 0.057) \times 10^{-3}$ to 0.024σ. The multiplicative coefficients $(9/4, 1)$ are flagged as *empirical*, not structurally derived; cross-K tests on $K \in \{\mathbb{Q}(i), \mathbb{Q}(\sqrt{-2}), \mathbb{Q}(\sqrt{-3}), \mathbb{Q}(\sqrt{-7}), \mathbb{Q}(\sqrt{-11})\}$ confirm $K = \mathbb{Q}(i)$ uniqueness. In a parallel modular-flavour direction, the CSD$(1+\sqrt{6})$ Littlest Modular Seesaw at $\tau_S = i$ (Ding–King–Liu–Lu Case B, [**?**, **?**]) yields a 2-parameter neutrino sector with NO, $m_1 = 0$, $\delta_{CP} \approx -87^\circ$ – testable by DUNE 2030+ at $\pm 15^\circ$. However, this prediction does *not* survive the W1 single-τ refit: Section **??** reports a hard 24.5σ pull on $\sin^2\theta_{13}$ that pushes the v7.4 picture toward a two-tau ($\tau_q \neq \tau_l$) reformulation. Section **??** records the G1.12.B campaign progress (M1+M2 PASS; M3-M6 forecast 3.5-5 month) which targets a parameter-free $B(p \to e^+\pi^0)/B(p \to K^+\bar{\nu}) \in [0.3, 3]$ Hyper-K/DUNE 2030+ falsifier. We discuss falsifiers (Tonks–Girardeau density, modular flavour lepton sector, NMC cosmological NULL at Cassini) and flag two open questions: the failure to cleanly fit $c = 7/10$ (tricritical Ising) and $c = 4/5$ (tetracritical Potts) under either bridge route, and the open $\sqrt{6}$ structure.

1 Setup: ECI v7.3 and the G1.15 refutation of H_7

The Extended Cosmological Index (ECI) research programme proposes that a single CM newform anchors both flavour-sector mass ratios (via S_4' modular flavour models [**?**, **?**, **?**]) and two-dimensional CFT Cardy density at minimal-model central charges, through the algebraic part of its critical L-values normalised by a power of the Chowla–Selberg period [**?**]. In the v7.3 axiomatic restart (committed in repository version 6.0.53, Zenodo DOI `10.5281/zenodo.20030684`), the relevant axioms were:

- H_5: *strict S-fixed point.* $\tau = i$ is the physical value of the modular parameter, motivated by maximal residual symmetry $\mathbb{Z}_4 \subset \mathrm{SL}(2,\mathbb{Z})$.
- H_7: *Damerell–Coates–Sinnott bridge.* The Cardy density $\rho_{\mathrm{Cardy}}(c) = c/12$ at minimal-model central charges is reproduced by the algebraic part of $L(f_{4.5.\mathrm{b.a}}, k/2)$ with $k = 5$, $f_{4.5.\mathrm{b.a}}$ the LMFDB newform 4.5.b.a (CM by $\mathbb{Q}(i)$).

The companion note [**?**] (V2) had already rendered H_5 suspect: it proves a no-go theorem for the Cabibbo angle in LYD20 Model VI at $\tau = i$ (numerical bound $\sin\theta_C \leq 0.005$ versus PDG 0.2253). The internal audit G1.15 (May 2026) then tested H_7 against the algebraicity hypotheses of Damerell [**?**], Shimura [**?**], and Deligne [**?**] for critical L-values of CM modular forms. These hypotheses require the critical point to be a non-negative integer (for the standard L-function of a weight-k CM form, the right-of-center critical points are $s \in \{k-1, k-2, \ldots, \lceil k/2 \rceil\}$; the parity of k controls which half-integer is included). For $k = 5$ odd, $s = k/2 = 5/2$ is *not* in the algebraicity domain proved by Damerell–Shimura nor in the period-extension of Deligne.

Proposition 1 (Refutation of H_7, G1.15 audit)**.** *The half-integer $s = 5/2$ is outside the algebraicity domain of Damerell [**?**] and Shimura [**?**] for a CM newform of weight $k = 5$ over $\mathbb{Q}(i)$. Consequently, the assertion $L(f_{4.5.b.a}, 5/2)\,\pi^{3/2}/\Omega_K^4 \in \mathbb{Q}$ is not entailed by any classical algebraicity theorem on which H_7 relied. The numerical value $L(f_{4.5.b.a}, 5/2) = 0.5200744676\ldots$ (LMFDB, mpmath dps= 60) is consistent with H_7 but not implied by any known theorem; absent such a theorem, H_7 is unfalsifiable in its v7.3 form and must be replaced.*

The audit log is recorded in repository commit 6.0.53; cf. Zenodo DOI `10.5281/zenodo.20030684`.

2 Sub-agent verdicts on the rescue

Three independent sub-agents (Sonnet harness, isolated worktrees) were tasked to explore rescue paths, all retaining $f_{4.5.\mathrm{b.a}}$ as the candidate CM anchor:

- **H7-A (POSITIVE).** Compute $L(f_{4.5.\mathrm{b.a}}, m)$ at the INTEGER critical points $m \in \{1,2,3,4\}$ via the Iwaniec–Kowalski approximate functional equation [**?**, Thm. 5.3] (mpmath dps= 60, sanity check $L(f_{4.5.\mathrm{b.a}}, 5/2) = 0.5200744676$ matching LMFDB). The algebraic parts $L(f_{4.5.\mathrm{b.a}}, m)\,\pi^{4-m}/\Omega_K^4$ form a rational ladder; in particular $1/12$ at $m = 2$ and $1/24$ at $m = 3$ *coincide exactly* with the Cardy densities $c/12$ at $c = 1$ (free boson) and $c = 1/2$ (Ising).
- **H7-B (POSITIVE, novel).** Perlmutter's degree-4 L-function attached to a 2D CFT torus partition function [**?**] matches the degree of the Rankin–Selberg square $L(f_{4.5.\mathrm{b.a}} \otimes f_{4.5.\mathrm{b.a}}, s)$, which has INTEGER critical points $s \in \{5, \ldots, 9\}$ and Petersson norm algebraic up to $\pi^{2k-1} = \pi^9$. This suggests an alternative bridge $\rho_{\mathrm{Cardy}} \propto L(f_{4.5.\mathrm{b.a}} \otimes f_{4.5.\mathrm{b.a}}, m_0)/\langle f_{4.5.\mathrm{b.a}}, f_{4.5.\mathrm{b.a}}\rangle$; we keep it as a backup since the explicit map is to be invented.

- **H7-C (STRUCTURAL + alternative).** A structural theorem of LYD20 ([**?**, Tab. Level4 MM], §3) shows that *every* hatted irrep of S_4' carries odd-weight forms, enforced by the action of the central element $R \sim -I$ in the metaplectic cover $\widetilde{\Gamma}(4)$ as $(-1)^k$. No even-weight CM-by-$\mathbb{Q}(i)$ form can substitute for $f_{4.5.\mathrm{b.a}}$ inside the $\widehat{\mathbf{3}}^{(5)}_{,2}$ embedding – a structural, not enumerative, obstruction. As an alternative central-point bridge, the Bertolini–Darmon–Prasanna anti-cyclotomic p-adic L-functions [**?**] *do* handle CM-by-imaginary-quadratic forms at half-integer central points; they preserve the original $s = k/2$ and replace the archimedean assertion by a p-adic one.

A scan of seven fermion observables (W1, PC compute, 2026-05-05) on a 30×30 grid in the τ-plane found a local minimum at

$$\tau^* = -0.1897 + 1.0034\,i, \qquad |\tau^* - i| = 0.19, \qquad \chi^2/\mathrm{dof} = 1.05, \tag{1}$$

which is a factor ≈ 3 closer to i than the LYD20 published best fit $\tau_{\mathrm{LYD20}} = -0.21 + 1.52\,i$ (with $|\tau_{\mathrm{LYD20}} - i| = 0.56$). This empirically supports *relaxing* H_5 from a strict fixed-point assertion to an *attractor* statement. Full synthesis at `notes/eci_v7_aspiration/H7_RESCUE/SYNTHESIS_2026-05-05.md` and verdict log at `notes/eci_v7_aspiration/W1_TAU_NEAR_I/results/W1_verdict_2026-05-05.log` in repository commit 6.0.53.

3 The amended axioms H_5' and H_7'

Axiom (H_5', CM-anchored attractor)**.** *The modular parameter τ of the S_4' flavour sector is not fixed at the S-fixed point $\tau = i$ but is a* CM-anchored attractor*: any joint fit to PDG fermion mass ratios and CKM/PMNS mixings should land within*

$$|\tau - i| \lesssim 0.2.$$

The empirical fit $\tau^* = -0.1897 + 1.0034\,i$ (W1 scan, $\chi^2/\mathrm{dof} = 1.05$, 7 observables) is the v7.4 reference point.

Remark 2. The relaxation $H_5 \to H_5'$ is forced by V2 [**?**]: at strict $\tau = i$, LYD20 Model VI's $\{d^c, s^c\}$-block of the down-quark matrix becomes near-rank-1 by an S-collinearity lemma, and $\sin\theta_C \leq 0.005$, a factor ≈ 45 below experiment. H_5' retains the CM anchoring (dynamical preference for the neighbourhood of $\tau = i$) while accommodating the experimentally required deviation.

Axiom (H_7', primary: integer-point Damerell ladder)**.** *For the LMFDB CM newform $f_{4.5.b.a}$ (CM by $\mathbb{Q}(i)$, weight $k = 5$, level $N = 4$) and the Chowla–Selberg period $\Omega_K = \Gamma(1/4)^2/(2\sqrt{2\pi}) \approx$ 2.62205755 for $\mathbb{Q}(i)$, the algebraic parts of the critical L-values at INTEGER critical points satisfy*

$$\frac{L(f_{4.5.b.a}, m)\,\pi^{4-m}}{\Omega_K^4} = \begin{cases} 1/10 & (m = 1), \\ 1/12 & (m = 2), \\ 1/24 & (m = 3), \\ 1/60 & (m = 4). \end{cases} \tag{2}$$

The Cardy density $\rho_{\mathrm{Cardy}}(c) = c/12$ at the unitary minimal-model central charges $c = 1$ (free boson) and $c = 1/2$ (Ising) is reproduced parameter-free *at $m = 2$ and $m = 3$ respectively.*

Axiom (H_7', complement: BDP anti-cyclotomic at $s = k/2$)**.** *At the half-integer central point $s = k/2$ outside the Damerell/Shimura/Deligne archimedean algebraicity domain, the appropriate algebraic bridge is the anti-cyclotomic p-adic L-function of Bertolini–Darmon–Prasanna [**?**] for the CM-by-imaginary-quadratic form. The p-adic value at $s = 5/2$ is algebraic in the Iwasawa algebra of the anti-cyclotomic $\mathbb{Z}_p$-extension of $\mathbb{Q}(i)$; we postulate (subject to A1 lit-check) a p-adic Cardy bridge for those CFT central charges that are* not *cleanly fit by the integer ladder of* (**??**)*, viz. $c \in \{7/10, 4/5\}$.*

4 Main numerical result and the Cardy hits

We collect the H7-A computation in tabular form (Table **??**). All values were obtained by mpmath (dps = 60) evaluation of the approximate functional equation [**?**, Thm. 5.3] for $L(f_{4.5.\mathrm{b.a}}, s)$, with the sanity-check $L(f_{4.5.\mathrm{b.a}}, 5/2) = 0.5200744676\ldots$ matching the LMFDB record exactly. The Chowla–Selberg period is $\Omega_K = \Gamma(1/4)^2/(2\sqrt{2\pi}) = 2.62205755\ldots$, the classical Lemniscate constant for $\mathbb{Q}(i)$ [**?**].

m	$L(f_{4.5.\mathrm{b.a}}, m)$	$L(f_{4.5.\mathrm{b.a}}, m)\,\pi^{4-m}/\Omega_K^4$	implied $c = 12\alpha_m$	CFT identification
1	$0.15244713359\ldots$	$1/10$	$6/5$	exotic, non-unitary minimal
2	$0.39910566245\ldots$	$\mathbf{1/12}$	$\mathbf{1}$	**free boson (Tonks–Girardeau)**
3	$0.62691370859\ldots$	$\mathbf{1/24}$	$\mathbf{1/2}$	**Ising**
4	$0.78780300053\ldots$	$1/60$	$1/5$	no clean unitary minimal CFT

Table 1: H7-A integer-point Damerell ladder for $f_{4.5.\mathrm{b.a}}$. Algebraic parts $\alpha_m := L(f_{4.5.\mathrm{b.a}}, m)\,\pi^{4-m}/\Omega_K^4$ are RATIONAL with denominators $\{10, 12, 24, 60\}$; the boxed entries reproduce $\rho_{\mathrm{Cardy}} = c/12$ at $c = 1$ and $c = 1/2$. The ladder is *standard*: $\alpha_1 = 1/10$ is the lemniscatic Hurwitz number H_4 (Hurwitz 1899 [**?**], prototype example in Harder–Schappacher 1986 [**?**] §1); $\alpha_2 = 1/12 = B_2/2 = -\zeta(-1)$ is the Bernoulli value transported to $\mathbb{Q}(i)$ via Eisenstein–Kronecker / Chowla–Selberg inside the Damerell–Shimura framework [**?**, **?**, **?**]. The functional equation $L(f_{4.5.\mathrm{b.a}}, s) \leftrightarrow L(f_{4.5.\mathrm{b.a}}, 5-s)$ forces $\alpha_m/\alpha_{5-m} = \Gamma(5-m)/\Gamma(m)$, hence $(\alpha_3, \alpha_4) = (\alpha_2/2, \alpha_1/6)$ are *not independent*; only (α_1, α_2) are.

Theorem 3 (Bernoulli-anchored Cardy hit)**.** *Within the Damerell–Shimura algebraicity framework, the algebraic part $\alpha_2 := L(f_{4.5.b.a}, 2)\,\pi^2/\Omega_K^4$ for $f_{4.5.b.a}$ = LMFDB `4.5.b.a` (CM by $\mathbb{Q}(i)$, weight 5) equals*

$$\alpha_2 \;=\; \tfrac{1}{12} \;=\; \tfrac{B_2}{2} \;=\; -\zeta(-1), \tag{3}$$

yielding the Cardy density $\rho_{\mathrm{Cardy}}(1) = c/12 = 1/12$ of the $c = 1$ free boson parameter-free*, where $B_2 = 1/6$ is the second Bernoulli number. The functional equation $L(f_{4.5.b.a}, s) = \varepsilon\, L(f_{4.5.b.a}, 5-s)$ then forces the companion $\alpha_3 = \Gamma(2)/\Gamma(3) \cdot \alpha_2 = 1/24$, which reproduces the $c = 1/2$ Ising central charge as a structural Γ-shadow of the same Bernoulli value.*

Proof. The Bernoulli identification $\alpha_2 = B_2/2$ for the lemniscatic CM-by-$\mathbb{Q}(i)$ family of weight ≥ 4 is the classical Eisenstein–Kronecker computation: at the integer critical point $m = k - 3 = 2$, the Damerell formula for a CM newform of weight k over an imaginary quadratic field K collapses to a Bernoulli rational times $\Omega_K^{2k-2}/\pi^{k-m-1}$; for $K = \mathbb{Q}(i)$ and $k = 5$ this is $B_2/2 \cdot \Omega_K^4/\pi^2$ [**?**, **?**, **?**]. The companion $\alpha_3 = 1/24$ then follows from the Γ-factor functional equation, with no independent input. □

Remark 4 (On the apparent "two parameter-free hits"). The empirical observation in Table **??** shows two Cardy-target rows ($c = 1$ at $m = 2$ and $c = 1/2$ at $m = 3$). These are *not* two independent algebraic coincidences: the Γ-factor functional equation *forces* $\alpha_3/\alpha_2 = \Gamma(2)/\Gamma(3) = 1/2$, so any α_2 yielding a clean c-value at $m = 2$ automatically produces a $c/2$ companion at $m = 3$. The genuine parameter-free coincidence is the $c = 1$ hit alone — a single Bernoulli identity $B_2/2 = 1/12$, with the Ising central charge appearing as its structural Γ-shadow. The strength of the bridge is correspondingly halved relative to the v7.3 prospectus, but it remains anchored on a textbook Eisenstein–Kronecker value.

Remark 5 (Two-step normalisation). The structure of (**??**) factorises as a Damerell-style binomial Ω_K^4/π^{4-m} times a small rational depending on m. The Ω_K^4 exponent is the k-th power expected in Shimura [**?**] for a weight-k CM form ($k = 5$ here, but the central L-values use $\Omega_K^{2(s-1)}$ with s critical; $s = m \in \{1, 2, 3, 4\}$ falls in the strip $1 \leq s \leq k-1$). The numerator pattern $(1/10, 1/12, 1/24, 1/60)$ is what requires literature anchoring – see Section **??**.

5 τ-plane probes I: the W1 attractor scan

To test whether τ near i is dynamically preferred – as opposed to merely allowed – we ran a 30×30 χ^2 scan on the domain $\mathrm{Re}\,\tau \in [-0.5, 0.5]$, $\mathrm{Im}\,\tau \in [0.7, 1.5]$ for seven fermion observables (m_c/m_t, m_u/m_c, m_e/m_μ, m_μ/m_τ, $\sin\theta_{12}$, $\sin\theta_{13}$, $\sin\theta_{23}$, all PDG 2024 central values). The scan ran on a single RTX 5060 Ti GPU (cosmopower-jax, 386 predictions/s) for 107 minutes.

	Best τ	$\lvert\tau - i\rvert$
LYD20 published [?]	$-0.21 + 1.52\,i$	0.56
W1 near-i minimum	$-0.1897 + 1.0034\,i$	**0.19**

Table 2: The W1 scan (PC compute, 2026-05-05) finds a local minimum $\tau^* = -0.1897 + 1.0034\,i$ with $\chi^2/\mathrm{dof} = 1.05$ (seven observables), a factor ≈ 3 closer to the S-fixed-point $\tau = i$ than the LYD20 best fit.

The seven observables fit the W1 minimum with $\chi^2_{\mathrm{up}} \sim 10^{-19}$, $\chi^2_{\mathrm{lep}} \sim 10^{-18}$, all χ^2 contribution coming from the CKM sector ($\chi^2_{\mathrm{ckm}} = 7.34$). This is consistent with H_5': *the empirical attractor lives at* $|\tau - i| \sim 0.2$, and the scaling is set by the same factor $\sqrt{\epsilon} \sim 0.063$ that governs the V2 no-go's near-rank-1 obstruction [?] – with $|\tau - i| \sim 0.2$ providing precisely the perturbation needed to break the S-collinearity.

Remark 6. The W1 scan does not yet include the CKM phase δ_{CP} nor the absolute neutrino mass scale; both are degrees of freedom not constrained by the seven observables used. A 64-observable refit including these will be the v7.4.1 update.

6 τ-plane probes II: $K = \mathbb{Q}(i)$ modular fixed-point and the Littlest Modular Seesaw (A14)

A second, independent probe of the $K = \mathbb{Q}(i)$ anchoring exploits the *strict* S-fixed point $\tau_S = i$ (rather than the W1 attractor τ^*) in the lepton sector. At $\tau_S = i$ the weight-2 S_4' triplet modular form $Y_3^{(2)}$ admits the alignment

$$Y_3^{(2)}(i) \;\propto\; \big(1,\; 1+\sqrt{6},\; 1-\sqrt{6}\big), \tag{4}$$

the so-called CSD($1+\sqrt{6}$) alignment of the Littlest Modular Seesaw (Ding–King–Liu–Lu Case B [?], King [?]). Equation (??) drives a 2-real-parameter neutrino sector with two right-handed Majorana singlets, predicting:

- **Hierarchy:** normal ordering (NO) with $m_1 = 0$ *forced* by the 2-RHν mechanism. Falsifier: JUNO 2030 NMO determination at 3–4σ.
- **Dirac CP phase:** $\delta_{CP} \approx -87^\circ$ at the CSD(3) benchmark; the CSD($1+\sqrt{6}$) value is close. **Falsifier:** DUNE 2030+ at $\pm 15^\circ$ rules out the prediction at $> 3\sigma$ if δ_{CP} centres at 0 or $+90^\circ$. Sensitivity sufficient by ~2032.
- $\sin^2\theta_{23}$: first octant, ≈ 0.46–0.55.
- $\Sigma m_\nu \approx 0.06\,\mathrm{eV}$ (the NO + $m_1 = 0$ floor), with $m_{\beta\beta} \sim$ few meV detectable by KamLAND-Zen 2027 / nEXO 2030+.

Remark 7 (Cross-K test confirms $K = \mathbb{Q}(i)$ uniqueness, A14). The DKLL19 Table 1 alignment lists for the five imaginary-quadratic CM points $K \in \{\mathbb{Q}(i), \mathbb{Q}(\sqrt{-2}), \mathbb{Q}(\sqrt{-3}), \mathbb{Q}(\sqrt{-7}), \mathbb{Q}(\sqrt{-11})\}$ show that *only* at $\tau_S = i$ ($K = \mathbb{Q}(i)$) does the weight-2 S_4' triplet $Y_3^{(2)}$ acquire the non-trivial

nested-radical alignment $(1, 1+\sqrt{6}, 1-\sqrt{6})$. At $\tau_{ST} = \omega$ ($K = \mathbb{Q}(\sqrt{-3})$) the alignment collapses to a permutation vector $(0, 1, 0)$, killing the seesaw. This K-uniqueness matches the $\alpha_2 = 1/12$ Bernoulli specificity of the Damerell ladder, and is the second independent $K = \mathbb{Q}(i)$ structural witness.

Remark 8 (A24: $\sqrt{6}$ is Galois-rational, NOT period-anchored). A PSLQ search at dps = 60, $|c| \leq 500$, in seven distinct bases $\{\sqrt{6}, \Gamma(1/4), \pi, \log 2, \log 3, \Omega_K, \log \theta_{2,3,4}(i), \log \eta(i)\}$ (A24) yields no Q-linear relation of $\log \sqrt{6}$ with the Chowla–Selberg periods of $\mathbb{Q}(i)$: in every non-trivial relation found, the coefficient on $\log \sqrt{6}$ is zero. The $\sqrt{6}$ in (**??**) is therefore a Galois-rational quantity (it is the discriminant $\sqrt{b^2 - 4c}$ of the secular 2×2 polynomial of $Y_3^{(2)}(i)$ eigenvalues), *not* a period-theoretic invariant of $\mathbb{Q}(i)$. Consequently the $K = \mathbb{Q}(i)$ anchor of CSD$(1 + \sqrt{6})$ is *combinatorial* (S-fixed-point of S_4'), not a deep CM-period identity via Chowla–Selberg. This keeps the empirical falsifier (DUNE 2030+) valid but classifies the $1 + \sqrt{6}$ alignment as a "thin coincidence" in the Lakatos sense, not a hard prediction lifted by deep number theory.

Empirical-deepness assessment. A14's $P \sim 20\%$ score ("WEAK CONSTRAINT, deep") reflects the combination: *pro* the structural cross-K test (DKLL19 Case B selects $K = \mathbb{Q}(i)$); *con* the absence of an explicit Ω_K in the modular-flavour formula and the independent fixed-point fit of Priya–Chauhan–Kumar–Nomura [**?**] which prefers $\tau_l = 3 + i/2$ rather than $\tau_l = i$. Section **??** below strengthens this *con* by orders of magnitude.

7 τ-plane probes III: numerological CKM alignments on the $K = \mathbb{Q}(i)$ Damerell ladder (A17)

A third, independent probe of $K = \mathbb{Q}(i)$ structure exploits the integer ladder of (**??**) *directly* on the CKM observables. A strict screen (small rationals q with $|\text{num}| + |\text{denom}| \leq 30$, $|\delta| < 0.5\%$) over all five imaginary-quadratic CM weight-5 newforms across $m \in \{1, 2, 3, 4\}$, $a \in \{0, \ldots, 5\}$ (the Damerell exponent), under the canonical $K = \mathbb{Q}(i)$ ladder yields two structurally clean candidates:

Candidate	Algebraic prediction	PDG / HFLAV measurement	Rel. error	Distance
$\lvert V_{us}\rvert$	$(9/4) \cdot \alpha_1 = 9/40 = 0.22500$	0.22501 ± 0.00068 (PDG 2024)	0.004%	$\mathbf{0.015\sigma}$
$\lvert V_{cb}\rvert^2$	$\alpha_1 \cdot \alpha_4 = 1/600 \approx 1.667 \times 10^{-3}$	$(1.668 \pm 0.057) \times 10^{-3}$ (HFLAV)	0.080%	$\mathbf{0.024\sigma}$

Table 3: A17 numerological CKM alignments on the canonical $K = \mathbb{Q}(i)$ Damerell ladder $\{\alpha_1, \alpha_2, \alpha_3, \alpha_4\} = \{1/10, 1/12, 1/24, 1/60\}$. Equivalently $|V_{cb}| = \sqrt{1/600} \approx 0.04082$ vs HFLAV 0.04085(7), 0.04σ off. No clean prediction emerged for $|V_{ub}|$ or $|V_{ub}/V_{cb}|$.

Remark 9 (Multiplicative coefficients are EMPIRICAL). The multiplicative coefficients $(9/4, 1)$ in Table **??** are *empirical*, not structurally derived. The match $|V_{us}| \approx 9/40$ is a restatement of the well-known Wolfenstein parameter coincidence $\lambda \approx 0.225$, now anchored to the Damerell $m = 1$ rational $\alpha_1 = 1/10$. The match $|V_{cb}|^2 \approx 1/600$ is the product of two known ladder rationals $\alpha_1 \cdot \alpha_4 = (1/10)(1/60)$, where the multiplicative coefficient 1 is empirically required. Without a derivation of (9/4) and (1) from a flavour-symmetry quantum-number argument, both rows of Table **??** should be regarded as *suggestive numerological alignments awaiting either group- theoretic interpretation or precision-experiment falsification.*

Remark 10 (Cross-K test confirms $K = \mathbb{Q}(i)$ uniqueness). Applying the same (m, a, q) choices to the analogous ladders of $K \in \{\mathbb{Q}(\sqrt{-2}), \mathbb{Q}(\sqrt{-3}), \mathbb{Q}(\sqrt{-7}), \mathbb{Q}(\sqrt{-11})\}$ produces $q \cdot r$ deviations from the same CKM target ranging from 524% to 15,490% (A17), confirming K-specificity of the alignments. The K-specificity is essentially a consequence of the Bernoulli/ Hurwitz alignment at $K = \mathbb{Q}(i)$ (already documented for $\alpha_2 = 1/12$), not independent SM evidence.

Falsifiers. The two CKM alignments of Table **??** are testable on near-future timescales:

- $|V_{us}| = 9/40 = 0.225000$: NA62 $K \to \pi\nu\bar{\nu}$ + FLAG-2027 lattice f_K/f_π should reach $\sim 0.3\%$ precision; survival margin ~ 0.5–1.5σ. **Testable 2027–2028.**
- $|V_{cb}|^2 = 1/600 = 1.667 \times 10^{-3}$: Belle II 2027 at $\sim 0.5\%$ on $|V_{cb}|^2$ gives $\sim 0.12\sigma$ separation (insufficient). LHCb-U2 / Belle II run 3 (2030+) at $\sim 0.2\%$ precision will discriminate.

A measured drift of $|V_{us}|$ outside 0.225 ± 0.001 at the 3σ level by 2028, or of $|V_{cb}|^2$ outside $1.667 \times 10^{-3} \pm 5 \times 10^{-6}$ at the 3σ level by 2030, falsifies the respective alignment.

8 Caveat: W1 τ-attractor does NOT accommodate the lepton sector at LYD20-style fits (A16)

The probes of Sections **??** (W1 quark+lepton joint fit at $\tau^* = -0.19 + 1.00\,i$, $\chi^2/\text{dof} = 1.05$), **??** (CSD$(1+\sqrt{6})$ at strict $\tau_S = i$ for the lepton sector), and **??** (Damerell-ladder CKM at $K = \mathbb{Q}(i)$) have been presented as if compatible. This subsection records that they are *not* compatible under a single-τ assumption: the W1 attractor τ^*, when imposed in the LYD20 unified-model lepton sector, gives a hard $\sim 24.5\sigma$ pull on $\sin^2\theta_{13}$ versus PDG/NuFIT 2024.

Fit	τ	χ^2_{min}	$\sin^2\theta_{13}$ pred.	Pull vs PDG
LYD20 published [**?**]	$-0.2123 + 1.5201\,i$	1567.6	0.02881	$+9.45\sigma$
W1 attractor	$-0.1897 + 1.0034\,i$	**883.0**	**0.00478**	$\mathbf{-24.53\sigma}$
Strict $\tau = i$	$0 + 1.0\,i$	2259.7	0.00735	-20.85σ
PDG/NuFIT 2024	—	—	0.02195(70)	—

Table 4: A16 single-τ refit of LYD20 unified-model lepton sector (arXiv:2006.10722) at three reference τ values. The W1 attractor τ^* that minimises χ^2 in the quark sector (Sec. **??**) under-predicts $\sin^2\theta_{13}$ by a factor ~ 4.6, a 24.5σ pull. Strict $\tau_S = i$ (the CSD$(1+\sqrt{6})$ point of Sec. **??**) is also off, by 20.85σ. The full fit JSON is at `notes/eci_v7_aspiration/A16_THETA13_PREDICTION/ theta13_prediction.json`.

Interpretation. The W1 attractor τ^* was located by minimising over seven *quark+lepton-mass-ratio* observables, NOT over the PMNS mixing angles directly. Including PMNS as constraints explicitly, the $\sin^2\theta_{13}$ shortfall is severe. Two non-exclusive readings:

- **(a) Two-tau reformulation required.** The flavour sector splits into two modular parameters (τ_q, τ_l) with $\tau_q \neq \tau_l$. τ_q remains in the W1 attractor basin near i; τ_l moves to a separate region (e.g. Priya–Chauhan–Kumar–Nomura's $\tau_l \approx 3 + i/2$ [**?**], or DKLL19 Case B's $\tau_l = i$ on a different rep multiplet) to reproduce $\sin^2\theta_{13} \approx 0.022$. This is the natural reading if Sec. **??**'s CSD$(1+\sqrt{6})$ is to be retained.
- **(b) LYD20 unified-model is too restrictive.** The six-parameter unified-model ansatz used in A16 may simply lack the rep-multiplet flexibility to reach $\sin^2\theta_{13} = 0.022$ at any τ near i; a Chen–King–Medina–Valle-style $\Gamma_4 \cong S_4$ ansatz ([**?**], non-GUT) might recover.

Operational consequence for v7.4. The single-τ axiom H_5' (Sec. **??**) must be weakened, in the lepton sector, to a *multi-tau* statement: either (i) H_5' applies to τ_q alone, with τ_l to be fitted independently in the next round (v7.4.1); or (ii) the LYD20 unified-model embedding must be replaced with one that has more freedom in the lepton sector while preserving the W1 quark-sector fit. Both options are open; a binary decision will be recorded in the v7.4.1 amendment after the planned 64-observable refit.

Remark 11 (Hallu #76 lineage). The single-τ assumption was inherited uncorrected from LYD20 itself, where the published best fit ($\tau = -0.21 + 1.52\,i$) reaches $\chi^2_{\rm min} = 1567$ over the same lepton observables. LYD20 reports the joint quark+lepton fit on a smaller observable set (e.g. omitting δ_{CP} and pinning $\sin^2\theta_{13}$ at its central value without σ-floor), which masks the pull. A16 uses the full PDG 2024 covariances and is a worse-case stress test.

9 G1.12.B campaign progress: M1+M2 PASS and the 2030+ proton-decay falsifier (A22, A26)

9.1 Note added in proof: G1.12.B M1+M2 PASS, M3-M6 forecast

The G1.12.B campaign [**?**] is the v7.4 proton-decay- falsifier programme: a six-milestone modular SU(5) $5_H + 45_H$ construction that, post-M_6, predicts a parameter-free branching ratio

$$\tfrac{B(p\to e^+\pi^0)}{B(p\to K^+\bar\nu)} \in [0.3,\,3] \quad (95\%\ \text{CL post-M}_6),$$

to be tested by Hyper-K + DUNE 2030+ data. Two of the six milestones are now closed:

- **M1 (A22) PASS 4/4.** The 45-of-SU(5) Higgs is assigned to the S_4' trivial singlet **1** at modular weight $k_{45} = 0$, with the off-diagonal Yukawa form-factor $f^{ij}(\tau)$ built from row-by-row LYD20-Model-VI contractions (rows u, c, t at weights $1, 2, 5$, irreps $\hat{3}', \mathbf{3}, \hat{3}$). The 3×3 matrix $f^{ij}(\tau = i)$ is computed symbolically (sympy) and numerically; all 6 off-diag entries are non-zero, modular invariance under S and T is verified to machine precision (residual $\leq 10^{-15}$), and the A2 +19.5% closure check matches at +19.43% (within 0.07 pp). The non-modular precedent for SU(5) $+45_H$ proton decay is Haba–Nagano–Shimizu–Yamada ([**?**], arXiv:2402.15124).
- **M2 (A26) PASS 5/5.** The off-diagonal up-quark Yukawa SVD at the W1 attractor $\tau^* = -0.1897 + 1.0034\,i$ reproduces the Antusch–Hinze–Saad GUT-scale targets [**?**] ($y_t = 0.4454$, $y_c/y_t = 2.725 \times 10^{-3}$, $y_u/y_c = 2.05 \times 10^{-3}$) to machine precision: the fit $(\beta_u/\alpha_u, \gamma_u/\alpha_u) = (152.42, 305.38)$ achieves $\chi^2_{\rm up} = 6.7 \times 10^{-20}$ (2 parameters, 2 observables, exact); $|\Delta(y_c/y_t)| = 10^{-9}\%$. U_L, U_R unitarity holds at $\sim 5 \times 10^{-16}$; SVD reconstruction at $\sim 7 \times 10^{-11}$.

The M3–M6 forecast is [**?**] 3.5–5 sub-agent months ($0–100): M3 is the full Patel–Shukla [**?**, Eq. (8)] 1-loop matching Wilson coefficients; M4 is a 2-loop SM RGE $\mathrm{M_{GUT}} \to M_Z$ refinement; M5 is the proton-decay partial widths $\Gamma(p \to e^+\pi^0, K^+\bar\nu, \mu^+K^0, \pi^+\bar\nu)$ via Haba 2402.15124 [**?**] + FLAG lattice matrix elements; M6 is a Bayesian scan in $(M_{T_5}, M_{T_{45}}, \eta, \kappa_{45})$ with an α_s unification prior, yielding the posterior on τ_p and the branching ratio. The decisive falsifier is:

> *ECI v7.4 modular SU(5)* $5_H + 45_H$ *at* $\tau \sim \tau^*$ *predicts* $B(p \to e^+\pi^0)/B(p \to K^+\bar\nu) \in [0.3,\,3]$ *(95% CL post-*M_6*). Hyper-K + DUNE combined 2030+ data outside* $[0.3,\,3]$ *at* $> 3\sigma$*, OR* $\tau(p \to e^+\pi^0) > 3 \times 10^{35}$ *yr at* $> 2\sigma \Rightarrow$ ***REFUTED***.

The current Super-Kamiokande limits are $\tau(p \to e^+\pi^0) > 2.4 \times 10^{34}$ yr [**?**], $\tau(p \to K^+\bar\nu) > 5.9 \times 10^{33}$ yr [**?**]. The G1.12.B forecast (post-M_6) is centred at $\tau(p \to e^+\pi^0) \sim 3 \times 10^{34}$ yr, $\tau(p \to K^+\bar\nu) \sim 10^{34}$ yr, both above the Super-K limits and well within the Hyper-K 20-yr [**?**] (10^{35} yr, 3×10^{34} yr) and DUNE 20-yr [**?**] (6.5×10^{34} yr) reach.

Anti-hallucination correction (hallu #78). Throughout the v7.3 codebase the SM running parameters of arXiv:2510.01312 had been internally tagged "Wang–Zhang 2025". The actual authors are **Antusch, Hinze, Saad (2025)** [**?**]; A18 caught the misattribution against the live arXiv API (2026-05-05). The numerical SM target values are unaffected (same paper); only the citation has been corrected. All in-text references in this v2 amendment use the correct authorship.

10 Open questions and pending verifications

Three independent open questions remain.

Q1: literature anchoring of the rational ladder — RESOLVED. The pattern $(1/10, 1/12, 1/24, 1/60)$ in Table **??** is the *standard* CM-by-$\mathbb{Q}(i)$ Eisenstein–Kronecker ladder for a weight-5 newform: $\alpha_1 = 1/10$ is the lemniscatic Hurwitz number H_4 first computed in Hurwitz 1899 [**?**] (§ III) as the prototype of the Deligne-conjecture instances later formalised by Damerell [**?**] and Shimura [**?**]; $\alpha_2 = 1/12$ is the second Bernoulli half-value $B_2/2 = -\zeta(-1)$ transported to $\mathbb{Q}(i)$ via Chowla–Selberg [**?**] and recorded as the prototype example in Harder–Schappacher 1986 [**?**] §1 (reproduced verbatim for $K = \mathbb{Q}(i)$ in Visser 2021 [**?**]). The functional equation reduces $(\alpha_3, \alpha_4) = (\alpha_2/2, \alpha_1/6)$ to dependent companions, so the ladder has only two independent rationals, both classical. H_7'(Damerell–ladder) is therefore a *theorem*, not a conjecture, with the strength restated as in Theorem **??** and the Remark following it.

Q2: the unfit central charges $c = 7/10$ and $c = 4/5$. The integer ladder (**??**) cleanly hits $c \in \{6/5, 1, 1/2, 1/5\}$ but *not* the unitary minimal central charges $c = 7/10$ (tricritical Ising) and $c = 4/5$ (tetracritical Potts). Two scenarios are possible:

- **(a)** The complement axiom H_7'(BDP) supplies these values via p-adic anti-cyclotomic L-values at $s = k/2$, with different small-prime contributions from the Iwasawa-algebra interpolation.
- **(b)** They genuinely require a *second* CM newform (the candidate would be the level-16 form 16.5.c.a, which appears in companion P-NT identifications of hatted weight-5 multiplets); in that case the v7.4 anchor must be extended to a two-form anchor.

Discriminating (a) from (b) is the next concrete test of v7.4.

Q3: Perlmutter degree-4 backup. Whether the Rankin–Selberg square $L(f_{4.5.\mathrm{b.a}} \otimes f_{4.5.\mathrm{b.a}}, s)$ at INTEGER critical points $s \in \{5, \ldots, 9\}$ supplies the same Cardy hits as the integer Damerell ladder is the H7-B research question; both routes should agree on $c = 1$ and $c = 1/2$ if either is the correct bridge.

Q4: $\sqrt{6}$ structure — partially CLOSED in the negative (A24). The $\sqrt{6}$ appearing in the CSD$(1+\sqrt{6})$ alignment of Eq. (**??**) is Galois-rational (discriminant of the secular polynomial of $Y_3^{(2)}(i)$ eigenvalues), *not* period-anchored via Chowla–Selberg (Remark **??**). PSLQ at dps= 60, $|c| \leq 500$ in seven distinct period bases finds no non-trivial $\mathbb{Q}$-linear relation $\log\sqrt{6} \leftrightarrow \{\log\Gamma(1/4), \log\pi, \log 2, \log\Omega_K, \log\theta_{2,3,4}(i), \log\eta(i)\}$. Future probes: $\mathbb{Q}(\sqrt{-24})/\mathbb{Q}(\sqrt{-6})$ period rings (which contain $\sqrt{6}$ natively), and the holomorphic $\tau = i + \epsilon$ deformation of Σm_ν.

11 Falsifiers

The v7.4 axiomatic restart, like its predecessors, must be falsifiable. Five concrete falsifiers are recorded.

1. **Tonks–Girardeau density measurement.** The Cardy formula at $c = 1$ predicts a one-loop free-boson density that, in the cold-atom Tonks–Girardeau limit, can be measured at the percent level. A deviation from $\rho = 1/12$ at the predicted operator at the 5% level falsifies H_7'(primary).

2. **Modular flavour lepton sector — A16 strengthens this.** The A16 fit (Sec. **??**) already pulls $\sin^2\theta_{13}$ at 24.5σ from PDG under the single-τ assumption with the LYD20 unified-model embedding. The decisive single-τ falsifier of H_5' is whether *any* embedding compatible with the W1 quark-sector fit reaches $\sin^2\theta_{13} = 0.022$ within $|\tau - i| \leq 0.3$. If the v7.4.1 64-observable refit fails this at $\geq 5\sigma$ for every embedding, H_5' is REFUTED in its single-τ form and must be reformulated as the two-tau ($\tau_q \neq \tau_l$) statement of Sec. **??**.

3. **NMC cosmological NULL at Cassini.** An independent branch of the ECI programme makes a non-minimally-coupled (NMC) scalar-field prediction for the γ-parameter of the Cassini Doppler tracking experiment (PPN bound $|\gamma - 1| \leq 2.3 \times 10^{-5}$). The prediction is a NULL: $|\gamma - 1| < 10^{-7}$ at present sensitivity, vanishing in the limit $\xi \to 0^+$. A future improvement of the Cassini bound to $\leq 10^{-6}$ that excludes the NMC NULL falsifies the cosmological branch of v7.4.

4. **DUNE 2030+ δ_{CP} (A14).** The CSD($1+\sqrt{6}$) Littlest Modular Seesaw at $\tau_S = i$ (Sec. **??**) predicts $\delta_{CP} \approx -87°$, NO with $m_1 = 0$, and first-octant $\sin^2\theta_{23} \in [0.46, 0.55]$. DUNE 2030+ at $\pm 15°$ on δ_{CP} rules out the prediction at $> 3\sigma$ if δ_{CP} centres near 0 or $+90°$. JUNO 2030 NMO at 3–4σ independently kills the NO + $m_1 = 0$ prediction if inverted.

5. **NA62 + FLAG-2027, LHCb-U2 / Belle II run 3 (2030+) on the Damerell-ladder CKM (A17).** The two numerological alignments of Table **??** are testable on near-future timescales: a measured $|V_{us}|$ outside 0.225 ± 0.001 at 3σ by 2028, or $|V_{cb}|^2$ outside $1.667 \times 10^{-3} \pm 5 \times 10^{-6}$ at 3σ by 2030, falsifies the respective alignment. These are weak (numerological) falsifiers compared to (1)–(4); they are listed for completeness.

6. **Hyper-K + DUNE 2030+ proton decay (G1.12.B, Sec. ??).** After M_6 closure (3.5–5 sub-agent months), the parameter-free branching ratio $B(p \to e^+\pi^0)/B(p \to K^+\bar{\nu}) \in [0.3, 3]$ is tested by Hyper-K + DUNE combined 2030+ data; outside [0.3, 3] at $> 3\sigma$, OR $\tau(p \to e^+\pi^0) > 3 \times 10^{35}$ yr at $> 2\sigma$, $\Rightarrow$ **REFUTED**.

12 Conclusion

The v7.4 amendment converts the strict fixed-point axiom H_5 into an *attractor* statement H_5' supported by a χ^2 scan (W1, 2026-05-05), and replaces the half-integer Damerell–CS bridge H_7 – refuted by the G1.15 audit on parity grounds – with a two-part bridge H_7' comprising (i) an INTEGER-point Damerell ladder (Table **??**) with parameter-free hits on $\rho_{\text{Cardy}} = c/12$ at $c = 1$ and $c = 1/2$, and (ii) a BDP p-adic anti-cyclotomic complement at the original central point. The CM anchor $f_{4.5.b.a}$ (LMFDB CM newform 4.5.b.a, CM by $\mathbb{Q}(i)$) is preserved across both rescue routes (H7-A, H7-B, H7-C all converge on this anchor).

Wave 4-5 follow-ups extend the v7.4 picture in three independent $K = \mathbb{Q}(i)$ probes: the CSD($1 + \sqrt{6}$) Littlest Modular Seesaw at $\tau_S = i$ (DUNE 2030+ δ_{CP} falsifier; A14 + A24); two numerological CKM alignments on the Damerell ladder ($|V_{us}| = 9/40$ to 0.015σ, $|V_{cb}|^2 = 1/600$ to 0.024σ, multiplicative coefficients flagged empirical; A17); and the G1.12.B campaign's parameter-free $B(p \to e^+\pi^0)/B(p \to K^+\bar{\nu}) \in [0.3, 3]$ Hyper-K/DUNE 2030+ falsifier (M1+M2 PASS, M3-M6 forecast 3.5-5 mo; A22 + A26).

A hard caveat (A16) is that the W1 attractor τ^* does NOT accommodate the lepton sector under the LYD20 unified-model embedding: $\sin^2\theta_{13}$ is pulled at 24.5σ. The single-τ axiom H_5' must therefore be weakened in the lepton sector, and the v7.4.1 update will record a binary decision between (a) two-tau reformulation $\tau_q \neq \tau_l$ and (b) replacement of the LYD20 unified-model embedding.

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Data and code availability

Repository: `https://github.com/AIdevsmartdata/crossed-cosmos`. Snapshot of v7.3 axioms: Zenodo DOI `10.5281/zenodo.20030684`(commit 6.0.53, 2026-05-05 06:00 UTC). H7-A computation: `notes/eci_v7_aspiration/H7_RESCUE/SYNTHESIS_2026-05-05.md`. W1 verdict log: `notes/eci_v7_aspiration/W1_TAU_NEAR_I/results/` `W1_verdict_2026-05-05.log`. Wave 4-5 follow-ups: A14 `A14_CM_NEUTRINO/SUMMARY.md`, A16 `A16_THETA13_PREDICTION/theta13_prediction.json` A17 `A17_PERIOD_RATIO_CKM/SUMMARY.md`, A22 `A22_G112B_M1/` (`f_ij_modular.py`), A24 `A24_SQRT6_STRUCTURE/` A26 `A26_G112B_M2/` (`m_u_svd.py`, `svd_results.json`).

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